A2 Problems: Fractions, ratios, percentages and powers

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No calculator for problems 1–8 and for the challenge problems, unless stated. A calculator is allowed for problems 9–17, but write down the calculation you type in. Give fractions in lowest terms.

Practice

1. Calculate.

(a) $\frac{2}{3}+\frac{1}{4}$ (b) $\frac{5}{6}-\frac{3}{8}$ (c) $2\frac{1}{3}-1\frac{3}{4}$ (d) $\frac{7}{9}\div\frac{14}{15}$ (e) $\frac{3}{4}\div\frac{2}{5}$ (f) $\frac{1}{2}+\frac{1}{3}+\frac{1}{6}$

2. (a) Write 0.375 and 0.45 as fractions.

(b) Write $\frac{5}{8}$ and $\frac{1}{7}$ as decimals.

(c) Write $0.\overline{27}$ and $0.1\overline{6}$ as fractions.

3. Without dividing, decide which of these give a terminating decimal. Explain each answer.

$$\frac{3}{40},\qquad \frac{7}{12},\qquad \frac{9}{75},\qquad \frac{11}{56},\qquad \frac{13}{125}$$

4. (a) Share €360 between three people in the ratio $2:3:4$.

(b) A recipe uses 300 g of pasta for 4 people. Use a ratio table (*verhoudingstabel*) to find how much pasta you need for 7 people.

(c) Concrete is mixed from cement, sand and gravel in the ratio $1:2:4$. How many kg of each are in 35 kg of concrete?

5. (a) What is 15% of 80?

(b) After a 20% discount a jacket costs €64. What was the original price?

(c) A price rises from €50 to €58. By what percentage?

(d) A price falls from €120 to €90. By what percentage?

6. Calculate, without a calculator.

(a) 2^{10} (b) $(-3)^3$ (c) $(-3)^4$ (d) -3^4 (e) 5^0 (f) $\left(\frac{2}{3}\right)^3$ (g) 10^{-2} (h) 2^{-3} (i) $\sqrt{196}$ (j) $\sqrt{0.16}$ (k) $\sqrt{\frac{9}{25}}$ (l) $\dfrac{3^7}{3^4}$

7. (a) Write in scientific notation (*wetenschappelijke notatie*): $72\,300\,000$ and $0.000\,058$.

(b) Calculate and give the answer in scientific notation:

(c) $(2.5\times10^6)\times(4\times10^{-3})$ (ii) $\dfrac{9\times10^5}{3\times10^{-2}}$ (iii) $(6\times10^4)^2$

8. Units and powers of 10.

(a) How many m² is 1 km²? Write it as a power of 10.

(b) How many litres is 1 m³? (1 litre = 1 dm³.)

(c) Write 5 mm² in m², in scientific notation.

School-exam level

9. A phone costs €400. Its price is raised by 20%, and later the new price is lowered by 20%.

(a) What is the final price?
(b) Would the answer change if the price were first lowered by 20% and then raised by 20%? Explain using growth factors.

10. A town has 50 000 inhabitants. The population grows by 3% per year.

(a) What is the growth factor per year?
(b) How many inhabitants are there after 5 years? Round to whole numbers.
(c) After how many whole years has the population first doubled? (Use your calculator to try values.)

11. You put €2000 in a savings account at 3% interest per year, and the interest is added to the account each year.

(a) How much is in the account after 5 years? Round to cents.
(b) How much would there be if you received 3% of the original €2000 each year, without interest on interest?

12. A bike costs €847 including 21% BTW (*btw*, VAT). What is the price excluding BTW, and how much BTW do you pay?

13. A walking map has scale $1 : 25\,000$.

(a) A path on the map is 6.4 cm long. How long is it in reality, in km?
(b) A lake on the map has area 3 cm². What is its real area in km²?

14. (a) The circumference of the Earth is about 4.0×10^7 m and light travels at about 3.0×10^8 m/s. How many times could light go round the Earth in one second?

(b) A human hair is about 8×10^{-5} m thick. How many hairs side by side make 1 cm?

15. You mix 200 ml of a drink that is 30% fruit juice with 300 ml of a drink that is 10% fruit juice. What percentage of the mixture is fruit juice? Why is the answer not 20%?

16. (a) Simplify $\sqrt{12} + \sqrt{27}$.

(b) Calculate $\sqrt{8} \cdot \sqrt{2}$ and $\dfrac{\sqrt{72}}{\sqrt{2}}$.

(c) Between which two whole numbers does $\sqrt{70}$ lie? Which is it closer to?
(d) Is $\sqrt{a^2} = a$ for every number a? Give an example or a reason.

17. Shop A offers 30% off. Shop B offers 20% off, and then a further 10% off the reduced price. Which shop is cheaper, and by what percentage of the original price?

Challenge

18. (a) Prove that $\sqrt{3}$ is irrational. (Hint: copy the proof for $\sqrt{2}$, using "divisible by 3" instead of "even". You'll need: if $3 \mid a^2$ then $3 \mid a$. Why is that true?)

(b) Where exactly does the same proof break down if you try it for $\sqrt{4}$?

19. Your friend says: "$0.999\ldots$ is a tiny bit less than 1." Give two different arguments that $0.\overline{9} = 1$. Then give a third: if two numbers are different, there is a number between them. What number lies between $0.\overline{9}$ and 1?

20. Prove: a fraction $\frac{a}{b}$ in lowest terms has a terminating decimal **if and only if** b has no prime factors other than 2 and 5. (Prove both directions. For "only if", use that prime factorisations are unique.)

21. (a) Find the decimals of $\frac{1}{7}, \frac{2}{7}, \ldots, \frac{6}{7}$. What do you notice?

(b) Find the length of the repeating block (the *period*) of $\frac{1}{13}$ and of $\frac{1}{17}$. (Long division by hand, or a calculator with care.)
(c) Explain why the period of $\frac{1}{n}$ is at most $n-1$.

22. No calculator. Which is larger?

(a) 2^{300} or 3^{200}
(b) 31^{11} or 17^{14} (Hint: compare both with a power of 2.)

23. How many digits does 2^{100} have? Prove your answer without a calculator, using $2^{10} = 1024$. (You may use that $1.024^{10} < 1.3$; can you prove that too?)

24. A shop raises a price by $p\%$ and later lowers the new price by $p\%$. The final price is 96% of the original. Find p.

25. An *Egyptian fraction* is a sum of different unit fractions $\frac{1}{n}$.

(a) Show that $\dfrac{1}{n} = \dfrac{1}{n+1} + \dfrac{1}{n(n+1)}$ for every whole number $n \geq 1$.
(b) Write $\frac{2}{7}$ as a sum of two different unit fractions.
(c) Write 1 as a sum of four different unit fractions.

26. The *mediant* of $\frac{a}{b}$ and $\frac{c}{d}$ (with a, b, c, d positive) is $\dfrac{a+c}{b+d}$.

(a) Find the mediant of $\frac{1}{2}$ and $\frac{2}{3}$ and check that it lies between them.
(b) Prove: if $\dfrac{a}{b} < \dfrac{c}{d}$ then $\dfrac{a}{b} < \dfrac{a+c}{b+d} < \dfrac{c}{d}$.
(c) Deduce that between any two different fractions there are infinitely many fractions.